

Nazia Tasnim

PhD. Student in Computer Science (September'23 - Present), Boston University.

Advisor: Dr. Bryan Plummer & Dr. Deepti Ghadiyaram

✉ nimzia@bu.edu

🌐 appledora.github.io

in [LinkedIn](#)

🎓 [Google Scholar](#)

📄 [appledora](#)

📍 Boston, MA.

RESEARCH INTEREST

I develop parameter-efficient methods for adapting deep neural networks with minimal trainable parameters. My research spans designing novel reparameterization and compression frameworks, developing evaluation protocols that expose failure modes in vision-language systems, and advancing mechanistic interpretability through weight-level analysis of knowledge representation. By unifying efficiency with interpretability, my work establishes foundations for controllable AI systems where model behavior can be precisely understood, adapted, and controlled at representation level.

SELECTED PUBLICATIONS

1. **Decompose, Mix, Adapt: A Unified Framework for Parameter-Efficient Neural Network Recombination and Compression** | IEEE/CVF Conference on CVPR'26
Nazia Tasnim*, Shrimai Prabhumoye & Bryan A. Plummer
 - Proposed a unified framework for model compression and parameter-efficient fine-tuning using shared basis matrices with gated mixer transformations, achieving **1.5%** improvement over SOTA PEFT with only **1.3K** trainable parameters per task
 - Demonstrated **+5.5%** improvement at **50%** compression ratio and **+2%** improvement at **75%** compression ratio on ViT-B/16 with **50x** distillation data efficiency, outperforming SOTA architectures and PEFT+MC combination methods VTAB-1K and other fine-grained benchmarks
2. **RECAST : Reparameterized, Compact weight Adaptation for Sequential Tasks** | ICLR'25
Nazia Tasnim* & Bryan A. Plummer
 - Proposed a novel reparameterization method that reduces the number of task-specific trainable parameters to fewer than **50**. We empirically demonstrated that RECAST outperforms the state-of-the-art by up to **3%** across various scales, architectures, and parameter spaces
3. **Seeing Isn't Orienting: A Cognitively Grounded Benchmark Reveals Systematic Orientation Failures in MLLMs** | ECCV'26
Nazia Tasnim* & Keanu Nichols et. al.
 - Developed a novel extensive benchmark to evaluate the object orientation perception of MLLMs, across **4** core dimensions encompassing **7** tasks curated from **11** datasets with **67** object categories across synthetic and real-world scenarios. The benchmark includes **33K+** Q&A pairs in MCQ format, revealing critical limitations of **22** closed/open-source VLMs.
4. **VISTA: Vision transformer enhanced by U-Net and image colorfulness frame filtration for automatic retail checkout** | IEEE/CVF Conference on CVPR-W'22
Nazia Tasnim* & Istiak Shihab et. al.
 - Implemented an end-to-end pipeline achieving **46%** accuracy in real-time multi-class product recognition, the **3rd** highest score for the cross-modality dataset.
5. **Vision-LLMs Can Fool Themselves with Self-Generated Typographic Attacks** | NeuRIPS-W'24
Maan Qraitem, Nazia Tasnim et. al.

- Introduced **two** novel self-generated attacks that prompt the LVLM to generate an attack against itself. Our benchmark reveals that Self-Generated attacks can reduce LVLM's classification performance by up to 33%
6. **Representation-Aware Unlearning via Activation Signatures: From Suppression to Knowledge-Signature Erasure** | EMNLP'26 Findings
Syed Naveed Mahmood, ..., **Nazia Tasnim**, et. al.
- Introduced Knowledge Immunization Framework (KIF), a representation-aware unlearning architecture that achieves near-oracle erasure ($FQ \approx 0.99$ vs. 1.00) while preserving utility at oracle levels ($MU = 0.62$), breaking the stability-erasure tradeoff and outperforming SOTA baselines including SimNPO by 54%
 - Developed dual-metric evaluation protocol exposing that standard foundation models exhibit scale-independent true erasure ($<3\%$ utility drift) across 3B-14B parameters, while reasoning-prior models reveal capacity-dependent transitions from behavioral suppression to genuine knowledge removal
7. **From Facts to Folklore: Evaluating Large Language Models on Bengali Cultural Knowledge** | ACL-IJCNLP'26
Nafis Chowdhury, ..., **Nazia Tasnim** et. al.
- Developed BLaNCK (Bengali Language Cultural Knowledge), a comprehensive evaluation framework with 5,265 terms from Wikipedia across 16 semantic categories including cultural (folk traditions, culinary arts, regional aspects) and non-cultural domains. The benchmark reveals that LLMs struggle significantly with Bengali cultural knowledge, with closed-source models outperforming open-source by large margins, and performance improving substantially when context is provided.
8. **OOD-Speech: A Large Bengali Speech Recognition Dataset for Out-of-Distribution Benchmarking** | INTERSPEECH'23
Fazle Rabbi Rakib, ..., **Nazia Tasnim** et. al.
- **1177.94** hours of audio data collected and curated from 22,645 native Bengali speakers from different regions, with additional **23.03** hours of speech collected and manually annotated from 17 OOD sources. This is jointly the largest and only OOD dataset for Bengali.

CURRENT ROLE

IVC-ML Group, Boston University

Sept'23 - Present

Graduate Researcher

- Investigating **weight-space decomposition for mechanistic interpretability** by applying sparse and atomic decomposition schemes to disentangle task-specific knowledge structures encoded in model parameters, enabling fine-grained analysis of how skills and concepts are distributed across weight components.
- Developed a **unified framework for compression and reparameterization** (CVPR'26) using shared basis matrices with gated mixer transformations, surpassing SOTA PEFT by 1.5+ points and SOTA compression by 5.5+ points at 50% compression ratio.
- Developed a **ultra-parameter efficient reparameterization scheme** (ICLR'25), for incremental learning that increases image classification performance by up to 3% with $< 2 * 10^{-6}$ tunable parameters.
- Developed an evaluation framework grounded in human-cognition development to isolate and assess **object orientation reasoning of LVLMs**
- Developed novel **probes and attacking techniques** (NeurIPS-W'24) to analyze vulnerabilities in LVLMs

EXPERIENCES

Adobe Inc. (San Jose, CA)

Summer'26

AI/ML Intern - Machine Learning Engineer, Digital Experience

- Develop two separate pipelines to process structured and freeform insights generated by Adobe Coworker
- Created a novel interpretable and controllable CJA insight reranking and diversification protocol that functions with extremely low latency

Boston University

Spring'25 and '26

Teaching Fellow

- Taught multiple lab sessions for two graduate-level courses with 100+ students, and supported the course instructors in planning and designing the structures of the course
- Designed, prepared, delivered, and graded all lab materials for **CS 505: Introduction to Natural Language Processing**. Currently doing the same for **CS598: Cyber Security and Entrepreneurship**
- Hosted discussions/office-hours with students, organized guest lectures and poster sessions, presented talks on research

Wikimedia Foundation

Sept'22 - June'23

Natural Language Processing Data Scientist

- Collaborated with Wikimedia's research team to **develop NLP tools** that support 300+ languages across 100+ wiki projects, with an *emphasis on efficiency and lowering resource requirements*
- Created research pipeline to **curate community resources** from wiki projects, developed assets through analysis, built models and off-the-shelf tools to be used in internal research

Giga Tech Limited

Aug'21 - June'22

Machine Learning Engineer

- Developed NLP submodules and pipelines for **Part-of-Speech (PoS), Named Entity Recognition (NER)** and **language model training** associated with the **Bangladesh National Syntactic Treebank** project
- Assisted in establishing annotation guidelines for diverse downstream tasks

Bengali.AI

Sept'21 - June'23

Research Affiliate

- **Coordinated teams, lead research projects** and published multiple research papers focused on alleviating the low-resource status of Bengali
- Launched **Google-funded Kaggle competitions** and inter-university DL contests with 600+ participants
- Developed some of the **largest benchmarking datasets for Bangla NLP**

SELECTED PROJECTS

mwtokenizer [\[Package\]](#)

A multilingual **Python tokenization package** for Wikimedia Projects focusing on non-whitespace delimited languages. Analyzed large-scale wikicorpora through **PySpark** to optimize pattern recognition in diverse writing systems and curated essential module assets.

mwparserfromhtml [[Package](#)]

A **Python** library to parse and extract metadata from Enterprise HTML Dumps. The module is part of the core components of the **Mediawiki Utilities Project**.

VISTA [[Code](#)]

Developed an end-to-end pipeline for real-time retail checkout, combining **UNET**-based segmentation with *entropy masking* for domain bias reduction. Implemented multi-class product classification using **Vision Transformers (ViT)**. Designed a **custom metric** for frame selection, optimizing object detection efficiency in video streams.

Shaako [[Code](#)] [[Demo](#)]

A mobile application to provide rural people with Emergency Medical Care support through community health workers. Built the frontend in **Android Framework**. The backend portal is **Django** and uses both **Firebase** and **Azure CMS** for database.

SUSTCast [[Code](#)] [[App](#)]

Created and launched the pioneering SUST campus online radio platform, featuring fully automated streaming via an **IceCast server** accommodating concurrent engagement of *2000+ listeners*. Designed the app interface using **Firebase** real-time database, while collaborating on the development of *data collection, processing, and music recommendation*.

HONORS AND AWARDS

- Recipient of **WiCV Mentorship Grant** | CVPR'25
- Recipient of **CRA GradCohort Fellowship** | Cohort 2025
- Recipient of **Dean's Fellowship**, Boston University GSAS. | Fall'2023 - Summer'2024
- **Outreachy Internship** at the Wikimedia Foundation (top 1.4% applicant) | Summer'22
- Second Runner-up at **Microsoft Imagine Cup** - SEANM | 2022
- Second Runner-up, **CVPR AI City Challenge** Track-4 (3rd globally) | 2022
- **Grace Hopper Scholar** | 2021

LEADERSHIP AND COMMUNITY SERVICE

- Research Instructor | **NSF REU Summer Program, 2025**
- Reviewer | **ICML, IJCV, ICLR, ECCV, CVPR, ACL-ARR**
- Judge | **BDOSN - NLP Hackathon, 2023**
- Founding Member & Membership Chair | **SUST ACM Students' Chapter, 2019-2020**
- Organizing Committee | **IEEE International Conference on Bangla Speech and Language Processing (ICBSLP), 2019**

TECHNICAL SKILLS

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|---------------------------|--|
| • Languages | Python, Java, JavaScript, SQL, SPARQL |
| • Framework | Android, Django, React, Flask |
| • Machine Learning | PyTorch, Keras, OpenCV, pySpark |
| • Research | Large-scale dataset curation, Controlled experiments and ablation, Literature review and Meta-analysis, Interdisciplinary research collaboration and communication |